

## 9. Implement a rule-based part-of-speech tagging system using regular expressions using python.

```
import nltk
from nltk.tag import RegexpTagger

patterns = [
    (r'.*ing$', 'VBG'),
    (r'.*ed$', 'VBD'),
    (r'.*ly$', 'RB'),
    (r'.*s$', 'NNS'),
    (r'.*', 'NN')
]

tagger = RegexpTagger(patterns)

sentence = "The boy is playing happily with friends"

words = sentence.split()

tags = tagger.tag(words)

print("Rule-Based POS Tags:")
print(tags)
```

... Rule-Based POS Tags:  
[('The', 'NN'), ('boy', 'NN'), ('is', 'NNS'), ('playing', 'VBG'), ('happily', 'RB'), ('with', 'NN')]